

Low Complexity ECG Biometric Authentication for IoT Edge Devices

Guoxin Wang, Deepu John, Avishek Nag
University College Dublin

Abstract—Wearable Internet of Things (IoT) devices are getting ubiquitous for continuous physiological data acquisition and health monitoring. This paper investigates an electrocardiogram (ECG) based biometric user authentication technique for IoT edge devices. A convolutional neural network (CNN) based deep learning technique for user authentication is proposed. The proposed technique achieves an authentication accuracy of 99.63% when tested with 290 subjects from Physionet PTB ECG database. To limit the complexity of the technique for IoT edge nodes, we applied optimisation techniques such as binarisation and approximation of the CNN weights. Accuracy-vs-time-complexity trade-off analysis is performed and results are presented for different optimisations. Our evaluations shows that the complexity-optimised method achieves 98.88% authentication accuracy with acceptable CPU cycles consumed.

Keywords—ECG, Authentication, Deep Learning, Low Complexity

I. INTRODUCTION

In the past few years, biometric authentication has been confirmed to have great application prospects in many fields [1]. With the proliferation of IoT wearable devices for data acquisition, physiological signals such as the electrocardiogram (ECG) can be continuously acquired in the background without requiring any specific user action. Therefore using ECG signal for biometric authentication enables the possibility of authenticating the user continuously. Compared with traditional discrete authentication methods, such as password and key card, continuous authentication systems are more secure as it prevents any intrusion after a one-time discrete authentication. ECG indicates the user's cardiac activity and is traditionally used for medical diagnosis purposes. ECG signal can be used for human authentication due to the following characteristics: (i) Universality(can be recorded from all humans) (ii) Distinguishability (no two ECG are exactly similar) (iii) Difficult to be forged.

In this paper, we present a convolution neural network (CNN) based technique which uses ECG for user authentication. To implement CNN type algorithms in IoT edge devices, we need to reduce computational complexity while maintaining high accuracy. Therefore, we propose using binary and approximate weights to replace original weights in the CNN. Finally the system performance is evaluated through simulations.

The remainder of this paper is organized as follows: Section II provides the details of system and low complexity improvements. The experimental results are presented in Section III and Section IV concludes the paper.

II. PROPOSED METHOD

This section introduces the deep-CNN based user authentication system.

A. Baseline System

The baseline system is a CNN model which is trained from an ECG dataset. This model is pre-trained with high accuracy for classification in different subjects. After some structure adjustments, we can use it for authentication. Firstly, we suppose some users as authenticated users and input ECG signals of these users into the model. The model will generate an output which can be used for user authentication and it will

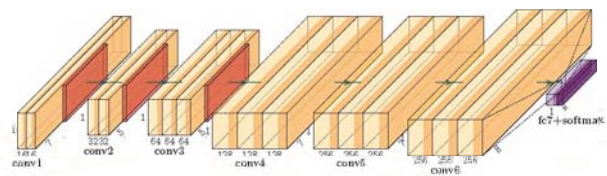


Fig. 1: Convolution Neural Network Structure

be stored into a database. Once a new user logs in, the model generate the output of the user, and it is used for calculating the Euclidean Distance between the output of new user and previous outputs stored in the database as a matching score. Finally, we decide if the new user passes the authentication by comparing the matching score with a threshold. The different steps of building this baseline system are discussed below.

1) Data Preprocessing

In this step, we extracted signal features from the original dataset. ECG signal contains several fiducial points which can be used to segment long signal into short pieces. Firstly, we re-sampled the signal to 200 Hz. Then, we detected all 'R' points and select a time window of 1 s for each 'R' point. Thus, we got many 'QRS' complexes each containing 200 samples. After that, we randomly selected 20 complexes. Once we complete the preprocessing of all subjects' records, those complexes can be used as input data for convolution neural network.

2) Modeling

We used a CNN structure inspired from [2] to build our model. To simplify the network structure and improve efficiency, we updated the network in [2] and adopted the structure shown in Figure 1. We used gradient descent to train the CNN model with 20 epochs. The batch size is 16.

3) Authentication

To generate high dimensional features of the ECG signals, we discarded fully-connected layers of trained CNN structure and kept the convolution layers which contain information of ECG signal. At first, we assumed some authenticated user had used their ECG signal to generate original database. Once we receive a new login request, our system will map it to the database then get a mapping score base on the Euclidean Distance:

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2} \quad (1)$$

where x_i -s are the high dimensional features' value of the tested data and y_i -s are the corresponding values stored in the database. If $d(x, y) \geq THRESHOLD$, the user passes the authentication and get the permission to login to the system, otherwise, the user will be rejected.

B. Low Complexity Improvements

CNN-based systems have excellent accuracy but high complexity since the weights in convolution layers are floating-point numbers. CNN output determination based on convolution calculations require many floating-point multiplications which influence performance. To deal with this issue, we used binary weights and approximate weights to replace the original weights.

1) Binary Weight

Binary weight method use binary numbers (1 and -1) to replace original floating point numbers. The *SIGN* function is applied to execute this transform:

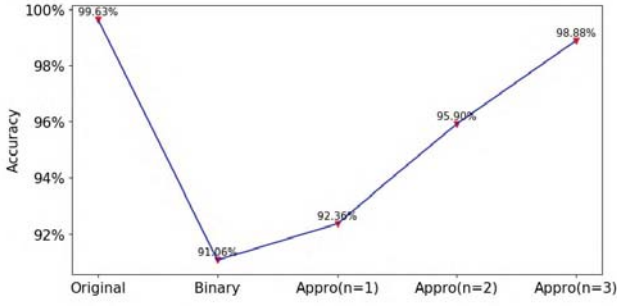


Fig. 2: Accuracy of Different Weight Variants

$$\bar{w}_i = \begin{cases} 1 & \text{if } w_i \geq 0 \\ -1 & \text{if } w_i < 0 \end{cases} \quad (2)$$

Compared with original weights, binary weights reduce the number of multiplications and only need to execute inversion operation in forward propagation.

2) Approximate Weight

In this method the original weights are approximated using exponential number [3]. Any floating-point weight a can be represented without loss of generality as $a = \frac{m}{2^n}$, where m is an integer. Further, it can be proven that any integer can be represented as the sum of a series of exponents ($m = 2^a + 2^b + \dots$). Hence, multiplications can be transformed to additions and bits-shift operations. n is a hyperparameter which controls the extent of approximation. As n increases the approximate weights become closer to the original weights.

III. EXPERIMENTAL RESULTS

A. Dataset

To train our model and evaluate performance of the system, we used PTB Diagnostic ECG Database [4]. The database contains 549 records from 290 subjects. Each subject is represented by one to five records. Each record includes 15 simultaneously-measured signals: the conventional 12 leads together with the 3 Frank lead ECGs. Each signal is digitized at 1000 samples per second, with 16-bit resolution over a range of ± 16.384 mV.

B. Evaluation

For the authentication system to be put into actual use, system accuracy is as important as time complexity. Hence, we evaluated the accuracy and response time of the system. We randomly divided the dataset into intruders and users for login test. For each login operation, we recorded the matching score. Finally, we gained total accuracy and counted the number of operations which can represent time complexity.

1) Authentication Accuracy

Figure 2 presents the accuracy of using different weight variants. The results show that the original weight has the highest accuracy (99.63%) because it uses weights that contain full information. Binary weight achieves the lower accuracy (91.06%) because during the processing from float number to binary number, the newly generated weight value is far from the original value, resulting in loss of information. Approximate weight has a better performance compared with binary weight and with higher n , greater accuracy is obtained.

2) Authentication Time Complexity

Table 1 presents the number of operations of using different weight variants which can represent time complexity.

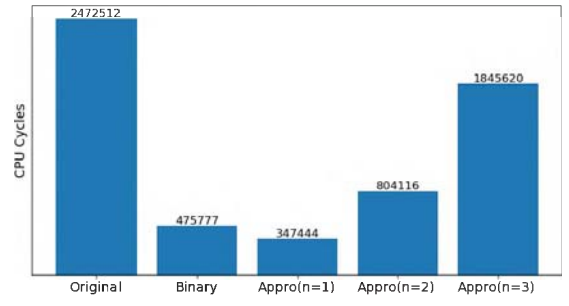


Fig. 3: CPU Cycles of Different Weight Variants

The results show that the original weight operates 823920 multiplications which results a large time complexity. Binary weight achieves the lowest time complexity because it only uses inversion which is a 1 bit operation. Approximate weight has a better performance compared with original weight because it uses bit shift and additions instead of multiplications. Increase in n leads to more bits to be shifted, increasing the run time.

TABLE I. Operations of Different Weight Variants

Weight Variants	Number of Operations			
	Multiplication	Inversion	Bit Shift	Addition
Original	823920	0	0	752
Binary	0	475025	0	752
Approximate (n=1)	0	81956	132744	132744
Approximate (n=2)	0	149496	327310	327310
Approximate (n=3)	0	266998	789311	789311

For processors with ‘AMD K7’ architecture, CPU clock cycles of various operations are as given : Multiplication: 3 cycles; inversion: 1 cycle; bit shift: 1 cycle; and addition: 1 cycle [5]. Figure 3 shows the exact running time (in terms of CPU cycles) for each weight variant, which prove binary-weight method and approximate-weight method obtained better time complexity than normal CNN weight.

IV. CONCLUSION

This paper introduced an ECG-based biometric authentication system which uses deep learning. It extracts QRS complexes from ECG dataset and uses them to train a high accuracy CNN model. This model can be optimised by changing its original weights to different weight variants for getting lower time complexity. To evaluate the performance of our system, we tested it using wide sets of samples acquired in uncontrolled conditions. Our system obtained excellent accuracy and results show that the proposed approach reduced the algorithm complexity.

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